

Health Insurance Charges Analysis

SQL Exploratory Analysis — Kaggle Health Insurance Dataset

1. Overview

This project explores a health insurance dataset sourced from Kaggle, using SQL in MySQL Workbench to identify the factors most strongly associated with insurance charges. The analysis covers smoking status, BMI, age, region, number of dependents, and sex.

2. Data Quality

- The dataset originally contained 1,338 records.
- A check for exact duplicate rows found 1 duplicate (0.07% of the data), which was removed, leaving 1,337 unique records.
- A check across all columns for missing or null values found none — the dataset was otherwise complete and required no further cleaning.

3. Key Finding: Smoking Status Drives Charges, and BMI Amplifies It

Smoking status is the single biggest driver of insurance charges. Smokers pay roughly 3-4x more on average than non-smokers across every BMI category. More importantly, the effect of BMI on charges depends heavily on smoking status: for non-smokers, BMI has almost no relationship with charges. For smokers, however, BMI has a strong relationship with charges — an obese smoker pays close to 5x what an obese non-smoker pays, and nearly double what a normal-weight smoker pays.

In other words, BMI alone does not drive up cost — it is the combination of smoking and high BMI that compounds charges dramatically, as shown below:

BMI Category	Non-Smoker Avg Charges	Smoker Avg Charges	Difference
Underweight	R5,533	R18,810	~3.4x
Normal	R7,686	R19,942	~2.6x
Overweight	R8,258	R22,496	~2.7x
Obese	R8,843	R41,558	~4.7x

4. Age

Age's relationship with charges only becomes clear once smoking status is held constant. Looked at across the whole dataset, age appears to have a weak relationship with charges. But within non-smokers and within smokers separately, older people consistently pay more than younger people in the same group. The reason age looked weak overall is that smoking status has such a large effect that it can override age when comparing across groups — for example, an older non-smoker can pay less than a younger smoker.

5. Region

Average charges differ by region, with the southeast having the highest average charges and the southwest and northwest the lowest. This is not fully explained by smoking rates — even when looking at

smokers only, the southeast still has the highest average charges of any region — suggesting region has a modest, independent effect on cost beyond smoking prevalence.

6. Number of Dependents and Sex

Neither the number of dependents (children) nor sex showed a meaningful relationship with charges. Having children did not correlate with higher or lower charges, and average charges were broadly similar between sexes.

7. Conclusion

Smoking status is the dominant factor in insurance charges, and its effect interacts with BMI — the combined presence of smoking and high BMI produces by far the highest charges in the dataset. Age has a real, positive relationship with charges within each smoking group, though it is masked when smokers and non-smokers are analyzed together. Region contributes a smaller, independent effect, while dependents and sex show little to no relationship with cost.